

Iván Cárdenas-León

Geospatial Specialist · PhD Candidate

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Library, ICT Services & Archive

Enschede, Netherlands

### Job Application for Research Data Steward

Dear Efe and LISA Team,

My name is Iván Cárdenas-León, I am currently in the last year of my PhD in Urban Digital Twins at the ITC faculty. Through this letter, I would like to apply for the position of Research Data Steward at LISA. I consider that my coding abilities coupled with my solution driven mindset and eight years of experience as an Environmental Engineer and researcher makes me a great candidate for this role.

In my current research I am developing **CrossTwin**, an open-source Urban Digital Twin platform across three urban sectors: drinking water supply, heat stress, and housing provision, which I expect turns into a software product and publication. Alongside this, I have built and deployed real operational tools: a Bibliographic reference enricher through CrossRef, Openalex and Semantic Scholar; a Streamlit web application for Circular Energy scenario analysis; a Python Panel monitoring dashboard for Vitens, the Dutch drinking water authority; and a ArcGIS Experience and geoprocessing tool for heat stress which allows for scenario-based modelling, impact assessment, and environmental cost analysis. For all of my research I have constantly used Git and GitHub for version control and collaboration. In my work I regularly use AI tools including Claude code and other LLM-assisted coding to accelerate prototyping and demo development. One of my publications includes a peer-reviewed Data paper as a result of a systematic literature review on the indicators used to monitor and evaluate impacts of planning decisions.

Beyond research, I have delivered applied geospatial solutions in consulting roles across Colombia and the Netherlands, including UAV imagery pipelines, Geo-database management with ETL pipelines and metadata management on environmental impact assessment projects, mining, energy transmission, flood risk modeling, Oil & Gas exploitation, and urban planning, and Web-GIS development using Mapbox GL and JavaScript. At Philips, I replaced a manual biodiversity spreadsheets workflow with an interactive global manufacturing site map integrating satellite and API-based land-use data. On both academic and consulting projects I have handled different data types and storage systems, including satellite imagery, databases, text files, spreadsheets and images with their corresponding metadata traceability and version control.

During the past three years, I have been designing and delivering university lectures on Digital Twins and Urban Heat Modelling for ITC and Vrije Universiteit Amsterdam, in these experiences I have had the opportunity to showcase my skills that reflect both my deep technical knowledge and my ability to communicate complex methods to diverse audiences.

I enjoy being able to apply my knowledge to ease the work of new researchers and to share my experience with them. I see the position of Research Data Steward as a great chance to explore opportunities within the research environment that do not necessarily translate into a research paper in itself, but are crucial for successful project management.

On the personal side, I have been living with my partner in Enschede for the past five years, and in my free time I play tycoon and strategy videogames, play beach volleyball, go hiking, find creative inspiration from illustrators, as of late, I am also been enjoying taking photographs of nature, animals and architecture.

I believe that my experience, skills and knowledge will be valuable for the researchers in the ET faculty and can contribute to the success of the LISA team.

Hoping to hear from you soon!

Sincerely,

**Iván Cárdenas-León**

PhD Candidate in Urban Planning, University of Twente

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